

Factors Influencing Likelihood of Death or Serious Injury In Motorcycle Riders

David Bean
University of Utah Department of Biomedical Informatics
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Abstract

Motorcycles have a reputation for being a far more dangerous method of transportation than other vehicles, but the severity of this increased danger is poorly understood by the general public. With the rapidly increasing cost of living in the United States, many commuters are looking for economical alternatives to larger vehicles in areas where public transportation infrastructure is lacking. Motorcycles are smaller, cheaper, and more fuel efficient than cars. The goal of this statistical analysis is to assess which factors may influence severity of motorcyclist injury and likelihood of death in the event of an accident. To this end, data from the United State's Crash Report Sampling System (CRSS) involving motorcycle accidents between the years 2016 and 2022 was analyzed. Several statistically significant, but relatively minor associations between motorcyclist safety decisions and risk of death or serious injury were uncovered. The most important indicators of serious injury were the consumption of alcohol or illicit substances by the motorcyclist prior to the accident. Two supervised machine learning models (Gaussian naive Bayes classifiers) were implemented in to predict injury severity or death based on independent variables from these motorcycle accident reports, with marginal success.

Introduction

Traffic congestion is an increasing issue in the United States. As the size, speed, and quantity of road vehicles increase every year, highway, parking lot, and surface road infrastructure fail to provide sufficient capacity. Public transportation has failed to provide a viable alternative to cars and trucks for commuting in many cities. The prioritization of street infrastructure for large passenger vehicles makes cities unwalkable. Bicyclists often must share lanes with distracted drivers commuting in multi-ton

SUVs and pickup trucks traveling several times their speed. Motorcycles can provide an economic and efficient method of transport. They are highly fuel efficient, generally capable of freeway speeds, and space compact. Three to four motorcycles can comfortably park in a single car parking spot.

Motorcycles may help the flow of traffic in states where lane splitting and lane filtering is allowed, reducing the distance traffic gets backed up at busy intersections.

Unfortunately, motorcyclists are at a significantly increased risk of serious injury and death when compared to car and light truck drivers. In 2022, 6,218 motorcyclists died on the road in the United States, making that year the most fatal for motorcyclists since 1975 despite major advancements in motorcycle safety technology [7]. Over the past decade, the number of deaths has continued to trend upward steadily. In 2021, the rate of fatalities for motorcyclists was 31.27 per 100 million miles traveled, and only 1.27 and 0.76 per 100 million miles traveled in cars and light trucks, respectively. That makes motorcyclists 24.6 times as likely to die than car drivers, and 41.1 times as likely to die than light truck drivers [7]. In the face of these daunting statistics, how does a potential motorcycle rider weigh the risks and benefits of buying a motorcycle for commuting? How does a current rider know which decisions they can make to improve their chances of avoiding serious injury and death on the road? This paper aims to answer these questions by analyzing motorcycle-related data from the Crash Report Sampling System (CRSS). The CRSS is a publicly accessible database maintained by the United States National Highway Traffic Safety Administration. It keeps detailed records of police vehicle accident reports that resulted in property damage, injury, or death from 60 counties across the United States [8].

This project aims to clarify connections between rider-controllable risk factors and injury severity in the event of a motorcycle accident by analyzing motorcycle accidents from the Crash Report Sampling System (CRSS). Descriptive statistics relating to injury severity outcomes were analyzed. Factors that showed the strongest correlations were used in the implementation of Gaussian Naive Bayes classifiers to develop a model that assesses a motorcyclist's chances of serious injury or death should they be involved in an accident based on factors within their control.

Methods

Data acquisition and preliminary processing Python was used for data analysis and visualization. The PANDAS Python package was used for data wrangling and processing, and the Matplotlib and Seaborn packages were used for data visualization. The CRSS database [8] was queried for all motorcycle accidents reported between 2016 and 2022. Each report contains observations on a variety of data elements from accidents. Each year has several CSV files containing data on a particular aspect of the accident report [5]. The files chosen to be included in this project were the ACCIDENT.csv, VEHICLE.csv, and the PERSON.csv files. The ACCIDENT.csv file contained general information about accident conditions that weren't directly related to any involved party. The VEHICLE.csv file contained information on vehicles involved in the accident, and the PERSON.csv file contained information on individuals involved. The CASENUM and VEH_NO (case number and vehicle number elements) were a composite key used to identify unique individuals in the merging of the person and vehicle datasets, and case number was used to merge in accident data. The resultant data set consisted of one line for each individual involved in any accident recorded in the CRSS between 2016 and 2022, regardless of the vehicle. This resulted in 873,784 records.

Statistical Analysis for Variable Selection The Crash Report Sampling System Analytical User's Manual [6] was used to identify data elements of interest and decode their variables across the three chosen data subsets. A number of variables were chosen for analysis and data harmonization. A summary of these variables and their purpose in this analysis can be found in *Table S3*.

Before the relation between injury severity and factors related to motorcycle accidents was assessed, the overall completeness of motorcycle accident reports was compared with the completeness of non-motorcycle accident report (*Figure 1a*), the report completeness by variable was observed in motorcycle vs non-motorcycle accidents (*Figure 1b*), and injury severity between motorcycle and non-motorcycle accidents was compared (*Figure 1c*).

The DRINKING variable, which indicated whether the vehicle operator was known to have been under the influence of alcohol at the time of the accident, and the DRUG variable, which indicated whether the vehicle operator was known to have been under the influence of an illicit substance at the time of the incident, had significantly lower completion than the other variables (*Figure 1a*). For this study, if the vehicle operator was not reported to be under the influence, it was presumed that they were sober. Missing values in these columns were imputed with 0, indicating sobriety. The preliminary selection of independent variables of motorcycle accidents was based on whether they were presumed to play a part in likelihood of serious injury. A correlation matrix was created to determine which of these elements had the strongest relationships amongst themselves (*Figure 2b*). In addition to the heat map, a scatterplot and 2D kernel density estimate matrix were made to help visualize any trends among quantitative elements in the data set (*Figure 2a*).

Analysis of Independent Variables An extensive series of hypothesis tests was performed to identify what, if any, statistically significant relationships existed between the chosen independent variables and injury severity or death. The data on injury severity (the dependent variable) is ordinal with 5 levels, and generally not normally distributed in relation to the independent variables (*Figure 3*). Therefore, non-parametric tests were utilized. Kruskal Wallis H tests were performed when three or more samples were compared. Mann-Whitney U tests were utilized when two samples were compared and for post-hoc analysis of the Kruskal Wallis tests. The α for all hypothesis tests was 0.05. The magnitude of relationships were measured using The Common-Language Effect Size (CLES) indicator. CLES effect size indicators provide an easily understandable description of the probability that a randomly chosen individual from one group will score higher than a randomly chosen individual from another group [4]. Additionally, Cliff's δ was used to examine effect size between samples with ordinal data. Reporting of Cliff's δ will be reserved for supplementary tables. Detailed descriptions of all hypothesis tests and their interpretations of their results can be found in *Table S1* and *Table S2*.

Bayesian Classifier Development The Python library Scikit-learn was used to develop the machine learning models. Those variables that showed a correlation with injury severity were used to train two Gaussian Naive Bayes classifiers. One classifier was trained to predict injury severity, and the other was trained to predict fatality. After filtering this dataset for report completeness of dependent variables, 9,704 motorcycle incident reports remained. The independent variables were: whether the rider was under the influence of alcohol, whether the rider was under the influence of drugs, the body class of motorcycle (three-wheeled/trike, moped/scooter, or standard motorcycle), the rider's reported gender, whether the rider was speeding, rider age, lighting conditions of the accident, the motorcycle's model year, the speed limit of the accident, and whether a helmet was used. These complete accident

records underwent 80:20 split (80% training and 20% testing). For the fatality prediction model, the dependent variable was whether the accident resulted in motorcyclist fatality: fatal or non-fatal. For the injury severity prediction model, the dependent variable was the rider's reported level of injury severity: uninjured, possible injury, minor injury, severe injury, fatal injury. The models were assessed for their ability to predict the dependent variables, and the importance of each variable in the model's prediction ability was determined.

Injury Severity (INJ_SEV)	Meaning
0	Uninjured
1	Suspected Minor Injury
2	Minor Injury
3	Severe Injury
4	Fatal Injury

Table 1) Dependent Variable (Injury Severity) Description table: The variable INJ_SEV encodes injury severity on a scale from 0 to 4. 0, being u, 4 being fatal.

Results and Discussion

Comparison of motorcycle to non-motorcycle accident reports. There were many more non-motorcycle ($n = 850,563$) accident reports in the CRSS dataset than motorcycle reports ($n = 23,221$), but the mean motorcycle report completeness was higher than non-motorcycle reports ($p < 0.05$, $CLES = 0.55$) and had a lower standard deviation of completeness ($SD\ 0.06 < SD\ 0.08$). The effect size of report completeness was small (*Figure 1a*). The injury severity of motorcycle accidents was much greater than non-motorcycle accidents ($p \approx 0$, $CLES = 0.88$). The most common injury severity in non-motorcycle vehicles was uninjured, which was the case for most non-motorcycle accident reports. The most common injury level for motorcycle accidents was minor injury. The

probability of an accident being fatal for motorcyclists in this dataset was 3.8%, whereas the probability of an accident being fatal for non-motorcycle vehicles was 0.52% (*Figure 1c*).

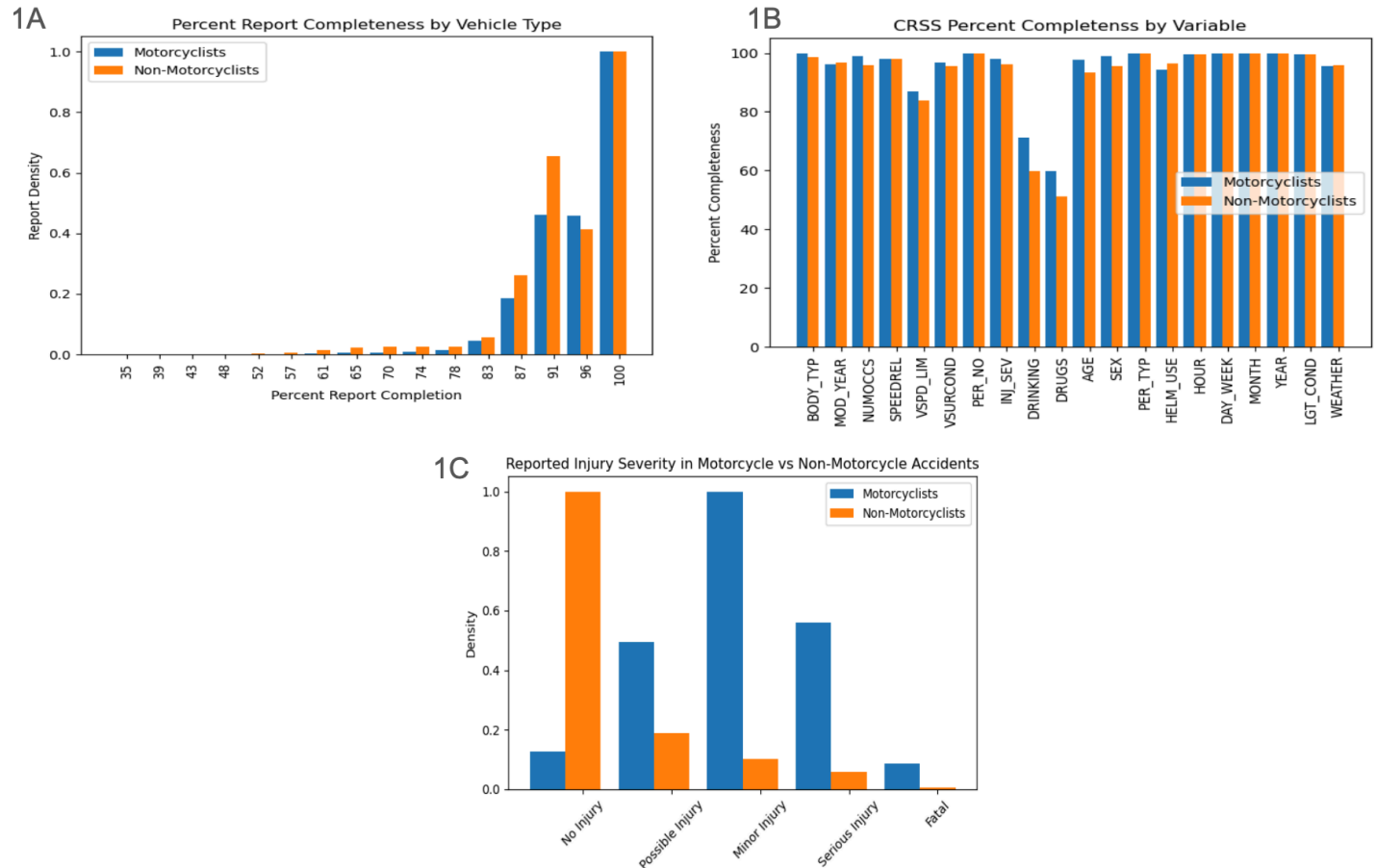


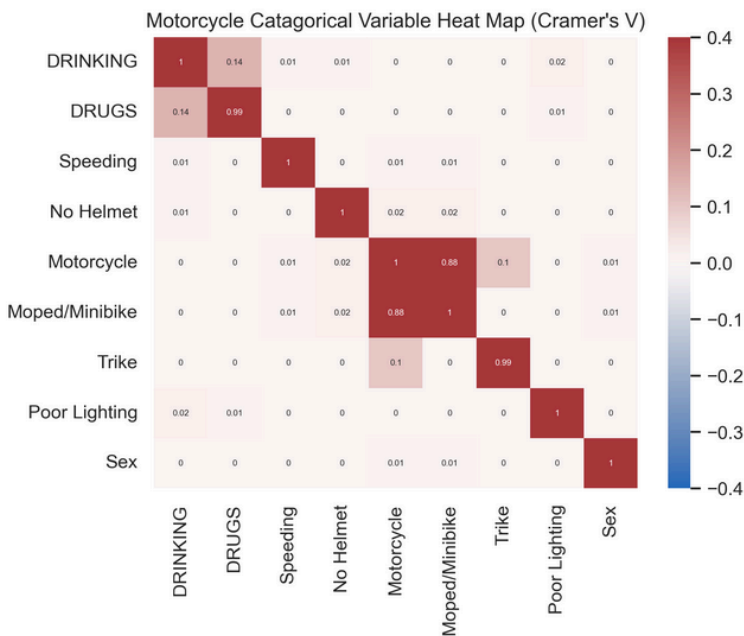
Figure 1) Comparison of Motorcycle to Non-Motorcycle accident reporting: Comparison of Motorcycle vs Non-Motorcycle accident reports. A) Percent report completeness by data element. B) Density or completeness levels for motorcycle-involved accidents vs non-motorcycle-involved accidents. C) Distribution of injury severity of motorcycles vs non-motorcycles.

Variables	Test	Null/Alt Hypotheses	Result	Interpretation and Associated Figure
Independent: Motorcycle or non-motorcycle accident report Dependent: Report Completeness in motorcycle accidents	Mann-Whitney U Test $n_{mc} = 23,221$ $n_{other} = 850,563$	H_0 Motorcycle Reports are not more complete than non-motorcycle reports H_a Motorcycle Reports are more complete than non-motorcycle reports	$U = 1.08 \times 10^9$ $p = 1.20 \times 10^{-170}$ Reject Null CLES: 55% Cliff's δ : 0.10	In 55% of cases, a given Motorcycle report completeness was higher than non-motorcycle report completeness. There is a negligible Cliff's δ effect size for report completeness by vehicle type. fig 1a
Independent: Motorcycle or non-motorcycle accident report Dependent: Injury Severity	Mann-Whitney U Test $n_{mc} = 10,599$ $n_{other} = 338,827$	H_0 Motorcycle and injuries are not more severe than other vehicle injuries H_a Motorcycle and injuries are more severe than other vehicle injuries	$U = 3.15 \times 10^9$ $p = \sim 0$ Reject Null CLES: 88% Cliff's δ : 0.75	In 88% of cases, a given motorcycle injury severity was higher than non-motorcycle injury severity. There is a large Cliff's δ effect size for injury severity by vehicle type. fig 1c

Table 2) Hypothesis testing for motorcycle to non-motorcycle comparisons. The difference in report completeness was statistically significant but rather small, but the difference in injury severity was large. Further hypothesis testing can be found in supplementary tables 1 and 2.

Analysis of motorcycle accident variables. Most of the relationships between variables in this data set were weak. Strength of correlation between categorical variables were assessed with Cramer's V scores [2]. Coefficients were rarely greater than 0. The strongest relevant correlation in this data was the relationship between drinking and drugs. When a motorcyclist was riding under the influence of alcohol, they were also more likely to be under the influence of an illicit substance ($V=0.14$) (Figure 2a). Relationships between continuous variables were also weak, and were difficult to discern (Figure 2b).

2A



2B

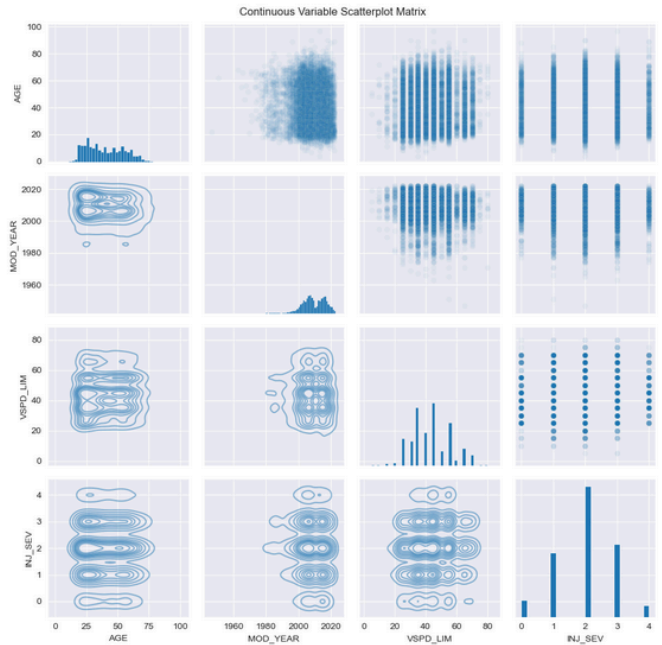


Figure 2) Analysis of Motorcycle Accident Variables: A) Heat map representing a correlation matrix of qualitative and quantitative variables related to motorcycle accidents in the CRSS dataset. Positive Cramer's V coefficients are encoded in red. The color scale is set between -0.04 and 0.04 to aid in the visualization of weak relationships. B) A scatterplot and KDE matrix of quantitative data elements of motorcycle accidents and their relationships.

Most categorical values played a small role in injury severity outcomes. Motorcyclist injuries tended to be worse under the conditions of speeding ($p = 9.37 \times 10^{-24}$), driving under the influence of alcohol ($p = 4.52 \times 10^{-29}$), and driving under the influence of drugs ($p = 1.12 \times 10^{-10}$). In 58% of cases, motorcycle injury severity was worse if the rider was speeding ($CLES = 0.58$) (Fig 3b). In 56% of cases, motorcycle accidents in poor lighting conditions resulted in more severe injury ($CLES = 0.53$) (Fig 3a). In 63% of cases, motorcycle accidents where the rider was drinking were more severe ($CLES = 0.63$) (Fig 3d), and the rate was nearly identical for motorcyclists who were under the influence of drugs ($CLES = 0.63$) (Fig 3d). Other categorical variables such as motorcycle body body class had very negligible effects on injury severity. The use of helmets had a small effect on injury outcomes, but riders without helmets were still worse off ($p = 1.28 \times 10^{-6}$ $CLES = 0.53$).

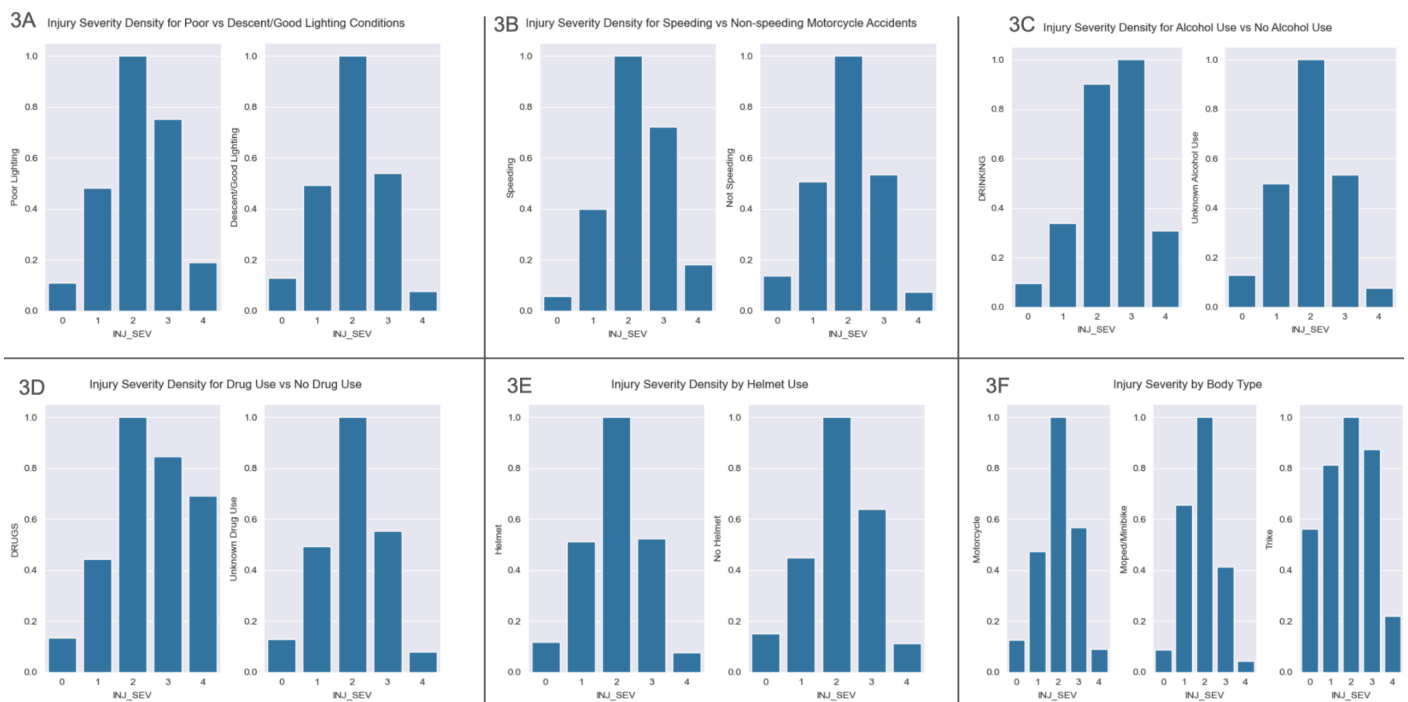


Figure 3) Distributions of Injury Severity for Categorical Variables: A) Distribution of injury severity for poor vs fair lighting conditions. B) Distribution of injury severity for speeding vs non-speeding accidents. C) Distribution of injury severity for alcohol vs non-alcohol involved accidents. D) Distribution of injury severity for drug vs non-drug involved accidents. E) Distribution of injury for accidents with and without helmet use. F) Distribution of injury severity for different motorcycle types.

Variables	Test	Null/Alt Hypotheses	Result	Interpretation and Associated Figure
Independent: Poor or fair lighting conditions Dependent: Injury Severity	Mann-Whitney U Test $n_{poor} = 1,006$ $n_{fair} = 9,593$	H_o Poorly lit motorcycle accidents are not more severe than well lit accidents H_a Poorly lit motorcycle accidents are more severe than well lit accidents	$U = 5.36 \times 10^6$ $p = 3.86 \times 10^{-10}$ Reject Null CLES: 56% Cliff's δ : 0.11	In 56% of cases, motorcycle accidents in poor light conditions were more severe. There was a small Cliff's δ effect size for lighting conditions. fig 3a
Independent: Whether the motorcyclist was under the influence of alcohol Dependent: Injury Severity	Mann-Whitney U Test $n_{drink} = 555$ $n_{nodrink} = 10,044$	H_o Motorcycle accidents involving alcohol are not more severe accidents without alcohol involvement H_a Motorcycle accidents involving alcohol are more severe accidents without alcohol involvement	$U = 3.51 \times 10^6$ $p = 4.52 \times 10^{-28}$ Reject Null CLES: 63% Cliff's δ : 0.26	In 63% of cases, motorcycle accidents involving alcohol were more severe. There was a small Cliff's δ effect size for alcohol involvement. fig 3c
Independent: Whether the motorcyclist was under the influence of drugs Dependent: Injury Severity	Mann-Whitney U Test $n_{drug} = 162$ $n_{nodrug} = 10,437$	H_o Motorcycle accidents involving drugs are not more severe accidents without drug involvement H_a Motorcycle accidents involving drugs are more severe accidents without drug involvement	$U = 1.07 \times 10^6$ $p = 1.12 \times 10^{-10}$ Reject Null CLES: 63% Cliff's δ : 0.11	In 63% of cases, motorcycle accidents involving drugs were more severe. There was a small Cliff's δ effect size for drug involvement. fig 3d
Independent: Speeding or non-speeding motorcycle accident Dependent: Injury severity	Mann-Whitney U Test $n_{speed} = 1,376$ $n_{nospd} = 9,223$	H_o Speed related motorcycle accidents are not more severe than non-speeding accidents H_a Speed related motorcycle accidents are more severe than non-speeding accidents	$U = 7.37 \times 10^6$ $p = 9.37 \times 10^{-25}$ Reject Null CLES: 58% Cliff's δ : 0.16	In 58% of cases, motorcycle injury severity was higher when the motorcyclist was speeding. There is a small Cliff's δ effect size for speeding. fig 3b

Table 3) Hypothesis testing for injury severity to categorical variable relations. The variables with the strongest relation to injury severity were alcohol use and drug use.

The continuous variables assessed in this analysis were rider age, the speed limit of the motorcycle at the time of the accident, and the model year of the motorcycle. A significant relationship was found among rider ages at different injury levels, with injuries tending to be slightly more severe with increased age ($p = 1.67 \times 10^{-3}$). Post-hoc analysis revealed that in 53% of cases, riders in fatal accidents were older than riders in accidents that resulted in possible injury ($p = 0.020$ CLES=0.53) and riders that resulted in minor injury ($p = 0.023$ CLES = 0.53) (Figure 4b). No significant relationship was found between ages of riders who died and ages of riders who were uninjured, possibly because the subset of accidents with these outcomes was very small in comparison to the others (Figure 4a).

Road speed limits showed similar trends to rider age, with statistically significant, but small effects.

Riders in fatal accidents were traveling on roads with higher speed limits than riders in accidents that

resulted in possible injury and riders that resulted in minor injury. Similar to age trends, no significant relationship was found between speed limits of riders who died and speed limits of riders who were uninjured (*Figure 4c*). Older model year motorcycles also tended to be associated with more severe injuries, with similar trends to the other two variables. Motorcycles involved in fatal accidents were older than motorcycles involved in accidents that resulted in possible injury and motorcycles involved in accidents that resulted in minor injury(*Figure 4e*).

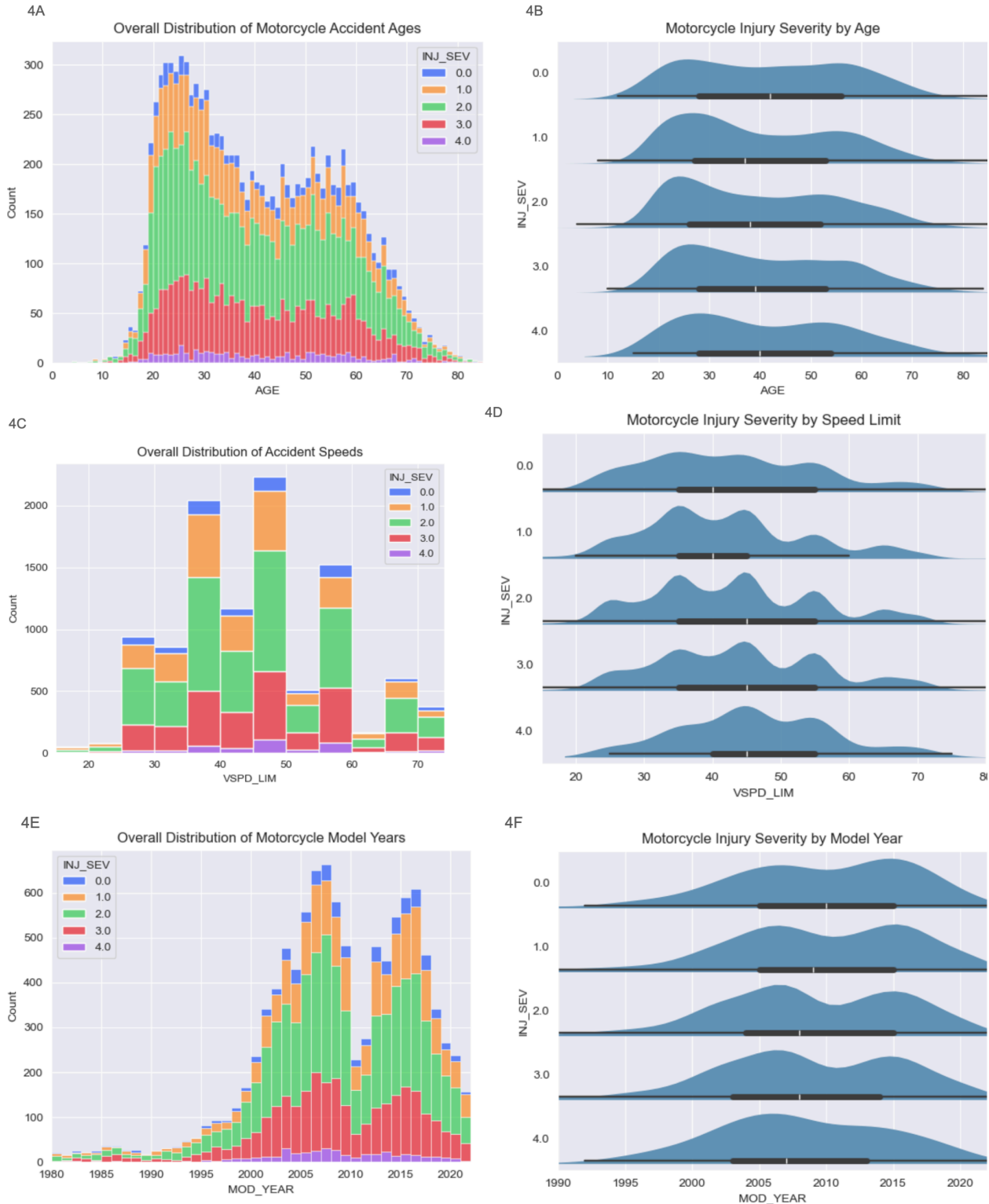


Figure 4) Distributions of Injury Severity for Continuous Variables: *A)* Overall distribution of motorcyclist ages. Severity levels are encoded with color. *B)* Distributions of Ages for different levels of injury severity. *C)* Overall distribution of motorcycle speed limits at location of accident *D)* Distribution of speed limits for different levels of injury severity *E)* Overall distribution of model years of motorcycles involved in accidents *F)* Distributions of model years for different levels of injury severity.

Modeling to Predict Motorcycle Accident Severity and Fatalities Two supervised machine learning algorithms were trained to the prediction of death in motorcycle accidents. Scikit-learn's Gaussian Naive Bayes classifier algorithm was used. The first classifier was trained to determine whether a rider lived or died. The accuracy of this model was 92%, the precision was 14%, and recall (the ability to identify fatal accidents) 22%. The weighted F1 score for this model was 0.9, an average dominated by the 96% of accidents that were not fatal. For fatal accidents alone, the F1 score was 0.17 (*Figure 5a*). The classifier for death prediction was heavily biased toward non-fatal accidents, likely because the vast majority of reports did not involve fatality, and those that did had very similar features to those with severe and minor injuries. Random chance and variables that weren't included in the training data play a significant role in whether someone will die in a motorcycle accident. Permutation importance of variables was performed on the model to identify the importance of each feature in the model's performance. The most important features were found to be drinking and drugs (*Figure 7a*).

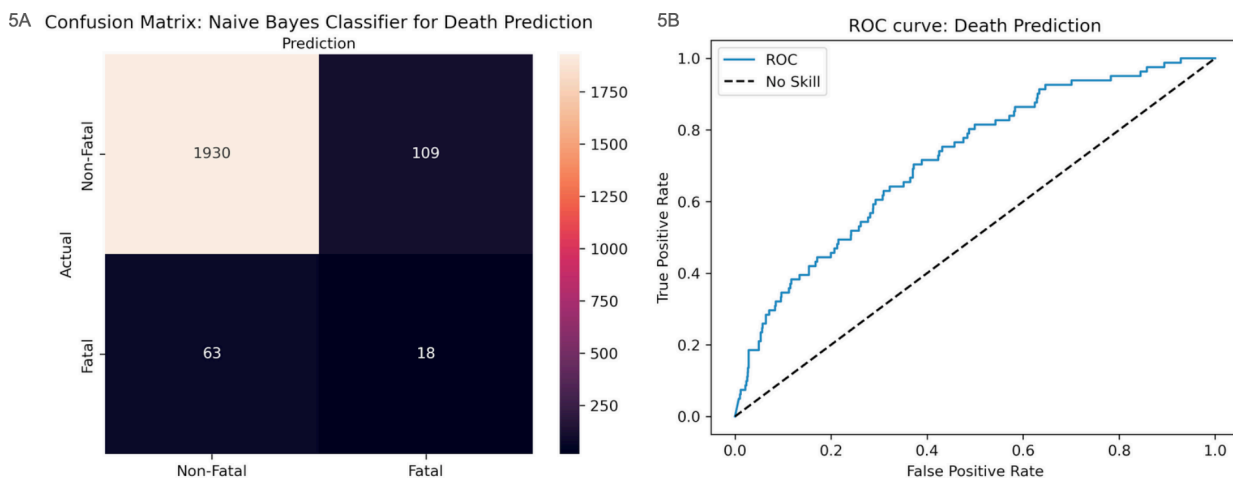


Figure 5) Confusion Matrices for Death-Predicting Supervised Models: A) Confusion Matrix for Gaussian Naive Bayes model of death prediction. B) ROC curve describing the performance of the model

A similar model was trained on the same data, this time with the level of injury severity as the target variable. This model had an accuracy score of 0.41 and was also heavily biased toward classifying injury levels as "Minor Injury" (*Figure 6*). Again, drinking was found to be the biggest predictor of injury severity, but the next largest predictor was vehicle body style (*Figure 7b*).

6 Confusion Matrix: Naive Bayes Classifier Injury Severity Prediction

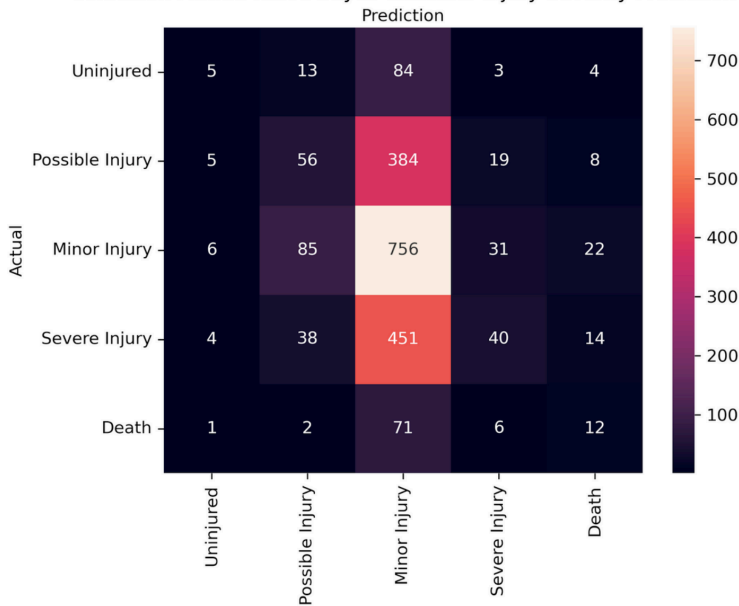


Figure 6) Confusion Matrix for Injury Severity-Predicting Model: Confusion Matrix for Gaussian Naive Bayes classifier for classification of injury severity. Diagonal from top left to bottom right are correct predictions

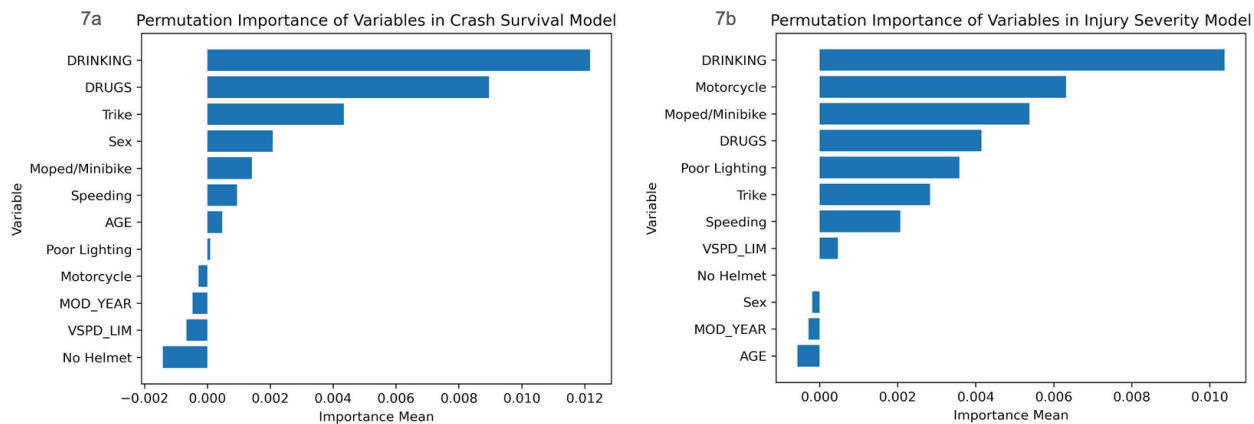


Figure 8) Permutation importance of variables for the Gaussian Naive Bayes classifiers. Bar length indicates the magnitude degradation in the model's performance when variable tampered with.

Conclusions

Motorcycle accident injuries were found to be significantly more severe than non-motorcycle accident injuries ($p \approx 0$, $CLES = 0.88$), with the most common injury level being minor, and the chance of death in the event of an accident being around 3.8% (*Figure 1c*). The strongest covariates with motorcycle accident injury and fatality outcomes were whether or not the driver had been drinking ($p = 4.52 \times 10^{-28}$ $CLES = 0.63$), whether they had been using drugs ($p = 1.12 \times 10^{-10}$ $CLES = 0.63$) and whether they had been speeding ($p = 9.37 \times 10^{-25}$ $CLES = 0.58$). The supervised machine learning models for predicting injury severity and death as a result of motorcycle accidents were heavily biased toward non-fatal accidents and minor injuries, as the vast majority of motorcycle accidents do not result in fatality, and the variables extracted from these accident reports have a large degree of overlap between the 5 levels of injury severity.

Although motorcyclists on average appear to still be at a significantly higher risk of death than non-motorcyclists, in the event of an accident, the factors that play the largest role in motorcycle accident severity in this analysis are entirely within the rider's control. It should be emphasized that the data analyzed here only contains information from motorcycle accidents that were severe enough to be reported, meaning that they likely involved injury or damage to another vehicle. Motorcyclists are unlikely to report simple incidents such as tip overs and short slides on secluded roads. Most motorcycle rides do not end in accidents. Although serious accidents may be unavoidable at times, riders can significantly mitigate risk of serious injury or death by remaining sober and following traffic laws.

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Independent: Whether the motorcyclist wore a helmet Dependent: Injury Severity	Mann-Whitney U Test $n_{nohelm} = 3,306$ $n_{helm} = 7,293$	H_o Motorcycle accidents without helmets are not more severe than accidents with helmets H_a Motorcycle accidents without helmets more severe than accidents with helmets	$U = 1.27 \times 10^7$ $p = 1.38 \times 10^{-6}$ Reject Null CLES: 53% Cliff's δ: 0.06	In 53% of cases, motorcycle accidents without helmet use were more severe than accidents with helmets. There was a negligible Cliff's δ effect size for helmet use. fig 3e
Independent: Motorcycle Body	Kruskal Wallis H Test	H_o There is no difference in median injury severity among	H = 35.2 p =	There is a significant difference in injury severity among the

type Dependent: Injury Severity		<i>the motorcycle body classes (motorcycle, moped, trike)</i>	2.31x10 ⁻⁸ Reject Null	motorcycle body types <i>fig 3f</i>
Injury Severity / Model Year	Kruskal Wallis H Test	<i>H_o</i> <i>There is no difference in motorcycle model year among injury severity levels</i>	H = 37.9 p = 1.19x10 ⁻⁷ Reject Null	There is a significant difference among model years of motorcyclists with given injury levels. <i>fig 4f</i>
Injury Severity / Speed Limit	Kruskal Wallis H Test	<i>H_o</i> <i>There is no difference in motorcycle speed limit among injury severity levels</i>	H = 80.9 p = 1.15x10 ⁻¹⁶ Reject Null	There is a significant difference among speed limits of motorcyclists with given injury levels. <i>fig 4d</i>
Injury Severity / Rider Age	Kruskal Wallis H Test	<i>H_o</i> <i>There is no difference in motorcyclist age among injury severity levels</i>	H = 17.4 p = 1.61x10 ⁻³ Reject Null	There is a significant difference among rider ages of motorcyclists with given injury levels. <i>fig 4a</i>

Table S1) Primary hypothesis testing

Post-Hoc Analysis	Tests	Null Hypotheses	Results	interpretation
Motorcycle Body Type Post-Hoc Analysis	Mann-Whitney U Tests $n_{motorcycle} = 9,369$ $n_{moped} = 920$ $n_{trike} = 111$	<u>Motorcycle to Moped</u> <i>H_o</i> <i>Motorcycle riders do not have more severe injuries than moped riders</i> <i>H_a</i> <i>Motorcycle riders have more severe injuries than moped riders</i> <u>Motorcycle to Trike</u> <i>H_o</i> <i>Motorcycle riders do not have more severe injuries than trike riders</i> <i>H_o</i> <i>Motorcycle riders do have more severe injuries than trike riders</i> <u>Moped to Trike</u> <i>H_o</i> <i>There is no difference in moped and trike injury severity</i> <i>H_a</i> <i>There is some difference in moped and trike injury severity</i>	<u>Motorcycle to Moped</u> U = 3.51x10 ⁶ p = 9.05x10 ⁻²⁸ Reject Null CLES: 63% Cliff's δ : 0.26 <u>Motorcycle To Trike</u> U = 3.96x10 ⁴ p = 6.38x10 ⁻⁷ Reject Null CLES: 64% Cliff's δ : 0.28 <u>Moped To Trike</u> U = 5.96x10 ³ p = 0.176 Fail to Reject Null CLES: 53% Cliff's δ : 0.07	<u>Motorcycle to Moped</u> 63% of motorcycle accidents are worse than moped accidents. Small effect size. <u>Motorcycle to Trike</u> 64% of motorcycle accidents are worse than trike accidents. Negligible effect size. <u>Motorcycle to Moped</u> 53% of moped accidents are worse than trike accidents. No effect size <i>fig 3f</i>
Motorcycle Model Year Fatality Post-Hoc Analysis	Mann-Whitney U Tests $n_{uninjured} = 597$ $n_{possible} = 2,307$ $n_{minor} = 4,679$ $n_{severe} = 2,611$ $n_{fatal} = 405$	<u>Fatal to Uninjured</u> <i>H_o</i> <i>There is no difference in model year between fatal and non-injury accidents</i> <i>H_a</i> <i>There is some difference in model year between fatal and non-injury accidents</i> <u>Fatal to Possible Injury</u> <i>H_o</i> <i>Fatal-injury accidents occur more often in older motorcycles than possible-injury accidents</i> <i>H_o</i> <i>Fatal-injury accidents do not occur more often in older motorcycles than possible-injury accidents</i>	<u>Fatal to Uninjured</u> U = 1.18x10 ³ p = 0.623 Fail to Reject Null CLES: 49% Cliff's δ : -0.02 <u>Fatal to Possible Injury</u> U = 4.97x10 ³ p = 0.020 Reject Null CLES: 53% Cliff's δ : 0.06	<u>Fatal to Uninjured</u> In 49% of cases, motorcycles in fatal accidents were older than motorcycles non-fatal accidents. No effect size <u>Fatal to Possible Injury</u> In 53% of cases, motorcycles in fatal accidents were older than motorcycles in possible-injury accidents. Negligible effect size

		<u>Fatal to Minor Injury</u> <i>H₀ Fatal-injury accidents occur more often in older motorcycles than minor-injury accidents</i> <i>H_a Fatal-injury accidents do not occur more often in older motorcycles than minor-injury accidents</i> <u>Fatal to Severe Injury</u> <i>H₀ There is no difference in model year between fatal and severe-injury accidents</i> <i>H_a There is some difference in model year between fatal and severe-injury accidents</i>	<u>Fatal to Minor Injury</u> $U = 1.00 \times 10^6$ $p = 0.023$ Reject Null CLES: 53% Cliff's δ: 0.06 <u>Fatal to Severe Injury</u> $U = 5.43 \times 10^5$ $p = 0.35$ Fail to Reject Null CLES: 0.51 Cliff's δ: 0.03	<u>Fatal to Minor Injury</u> In 53% of cases, motorcycles in fatal accidents were older than motorcycles in possible-injury accidents. Negligible effect size <u>Fatal to Severe Injury</u> In 51% of cases, motorcycles in fatal accidents were older than motorcycles non-fatal accidents. No effect size <i>fig 4f</i>
Motorcycle Fatality Age Comparisons Post-Hoc Analysis	Mann-Whitney U Tests $n_{uninjured} = 597$ $n_{possible} = 2,307$ $n_{minor} = 4,679$ $n_{severe} = 2,611$ $n_{fatal} = 405$	<u>Fatal to Uninjured</u> <i>H₀ There is no difference in rider age between fatal and non-injury accidents</i> <i>H_a There is some difference in rider age between fatal and non-injury accidents</i> <u>Fatal to Possible Injury</u> <i>H₀ Fatal-injury accidents occur more often in older riders than possible-injury accidents</i> <i>H₀ Fatal-injury accidents do not occur more often in older motorcycles than possible-injury accidents</i> <u>Fatal to Minor Injury</u> <i>H₀ Fatal-injury accidents occur more often in older riders than minor-injury accidents</i> <i>H₀ Fatal-injury accidents do not occur more often in older motorcycles than minor-injury accidents</i> <u>Fatal to Severe Injury</u> <i>H₀ There is no difference in rider age between fatal and non-injury accidents</i> <i>H_a There is some difference in rider age between fatal and severe-injury accidents</i>	<u>Fatal to Uninjured</u> $U = 1.18 \times 10^6$ $p = 0.62$ Fail to Reject Null CLES: 49% Cliff's δ: -0.02 <u>Fatal to Possible Injury</u> $U = 4.97 \times 10^5$ $p = 0.020$ Reject Null CLES: 53% Cliff's δ: 0.07 <u>Fatal to Minor Injury</u> $U = 1.00 \times 10^6$ $p = 0.023$ Reject Null CLES: 53% Cliff's δ: 0.06 <u>Fatal to Severe Injury</u> $U = 5.44 \times 10^5$ $p = 0.36$ Fail to Reject Null CLES: 51% Cliff's δ: 0.03	<u>Fatal to Uninjured</u> In 49% of cases, riders in fatal accidents were older than riders in non-fatal accidents. No effect size <u>Fatal to Possible Injury</u> In 53% of cases, riders in fatal accidents were older than riders in possible-injury accidents. Negligible effect size <u>Fatal to Minor Injury</u> In 53% of cases, riders in fatal accidents were older than riders in possible-injury accidents. Negligible effect size <u>Fatal to Severe Injury</u> In 51% of cases, riders in fatal accidents were older than riders in severe-injury accidents. No effect size <i>fig 4b</i>
Motorcycle Fatality Speed Limit Comparisons Post-Hoc Analysis	Mann-Whitney U Tests $n_{uninjured} = 597$ $n_{possible} = 2,307$ $n_{minor} = 4,679$ $n_{severe} = 2,611$ $n_{fatal} = 405$	<u>Fatal to Uninjured</u> <i>H₀ There is no difference in speed limit between fatal and non-injury accidents</i> <i>H_a There is some difference in speed limit between fatal and non-injury accidents</i> <u>Fatal to Possible Injury</u> <i>H₀ Fatal-injury accidents do not occur at higher speeds than possible-injury accidents</i>	<u>Fatal to Uninjured</u> $U = 1.18 \times 10^5$ $p = 0.62$ Fail to Reject Null CLES: 49% Cliff's δ: -0.02 <u>Fatal to Possible Injury</u> $U = 4.97 \times 10^5$ $p = 0.02$ Reject Null	<u>Fatal to Uninjured</u> In 49% of cases, speed limits in fatal accidents were higher than riders in non-fatal accidents. No effect size <u>Fatal to Possible Injury</u> In 53% of cases, riders in fatal accidents were older than riders in possible-injury

		<i>H₀ Fatal-injury accidents do occur at higher speeds than possible-injury accidents</i> Fatal to Minor Injury <i>H₀ Fatal-injury accidents do not occur at higher speeds than minor-injury accidents</i> <i>H₀ Fatal-injury accidents do occur at higher speeds than minor-injury accidents</i> Fatal to Severe Injury <i>H₀ There is no difference in speed limit between fatal and severe-injury accidents</i> <i>H_a There is some difference in speed limit between fatal and severe-injury accidents</i>	CLES: 53% Cliff's δ: 0.06 Fatal to Minor Injury U = 1.00x10⁶ p = 0.02 Reject Null CLES: 53% Cliff's δ: 0.06 Fatal to Severe Injury U = 5.43x10⁵ p = 0.35 Fail to Reject Null CLES: 51% Cliff's δ: 0.02	accidents. Negligible effect size Fatal to Minor Injury In 53% of cases, riders in fatal accidents were older than riders in possible-injury accidents. Negligible effect size Fatal to Uninjured In 49% of cases, speed limits in fatal accidents were higher than riders in non-fatal accidents. No effect size
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Table S2) Post-hoc analysis for kruskal-wallis tests

Variable Name	Description	Location	Type	Use
CASENUM	Unique identifier for each incident report	ACCIDENT.csv, PERSON.csv, VEHICLE.csv	Identifier	Harmonization of datasets
VEH_NO	Unique identifier for each vehicle involved in a given report	VEHICLE.csv, PERSON.csv	Identifier	Harmonization of datasets
BODY_TYP	Body type description of vehicles	VEHICLE.csv	Qualitative Nominal	Independent variable
MOD_YEAR	Vehicle model year	VEHICLE.csv	Qualitative Ordinal	Independent variable
NUMOCCS	Number of occupants in vehicle	VEHICLE.csv	Quantitative Discrete	Independent variable
SPEEDREL	Whether the vehicle was speeding or not	VEHICLE.csv	Qualitative Nominal	Independent Variable
VSPD_LIM	Legal speed limit of vehicle in question	VEHICLE.csv	Quantitative Continuous	Independent Variable
VSURCOND	Road surface condition at location of accident	VEHICLE.csv	Qualitative Nominal	Independent Variable
HOURL	Hour of day of accident	ACCIDENT.csv	Quantitative Discrete	Independent Variable
DAY_WEEK	Day of the week of incident	ACCIDENT.csv	Quantitative Discrete	Independent Variable
YEAR	Year of incident	ACCIDENT.csv	Quantitative Discrete	Independent Variable
LGHT_COND	lighting conditions of accident	ACCIDENT.csv	Qualitative Nominal	Independent Variable
WEATHER	Weather conditions at time of incident	ACCIDENT.csv	Qualitative Nominal	Independent Variable
PER_NO	Identifier of person involved in each accident	PERSON.csv	Qualitative Nominal	Data Harmonization
DRINKING	Whether person was known to be under the influence of alcohol	PERSON.csv	Qualitative Nominal	Independent Variable
DRUGS	Whether person was known to be under the influence of drugs	PERSON.csv	Qualitative Nominal	Independent Variable
AGE	Age of person	PERSON.csv	Quantitative Discrete	Independent Variable
PER_TYP	Whether the the vehicle operator or a	PERSON.csv	Qualitative Nominal	Independent Variable

	passenger			
HELM_USE	Whether the person was wearing a seat belt (for cars) or a helmet (for motorcycles)	PERSON.csv	Quantitative Nominal	Independent Variable
INJ_SEV	Severity of Injury (or death) of person	PERSON.csv	Quantitative Ordinal	Dependent Variable

Table S3) Variable names, origins, and definitions